

10.22 Functional Decomposition

10.22.1 Purpose

Functional decomposition helps manage complexity and reduce uncertainty by breaking down processes, systems, functional areas, or deliverables into their simpler constituent parts and allowing each part to be analyzed independently.

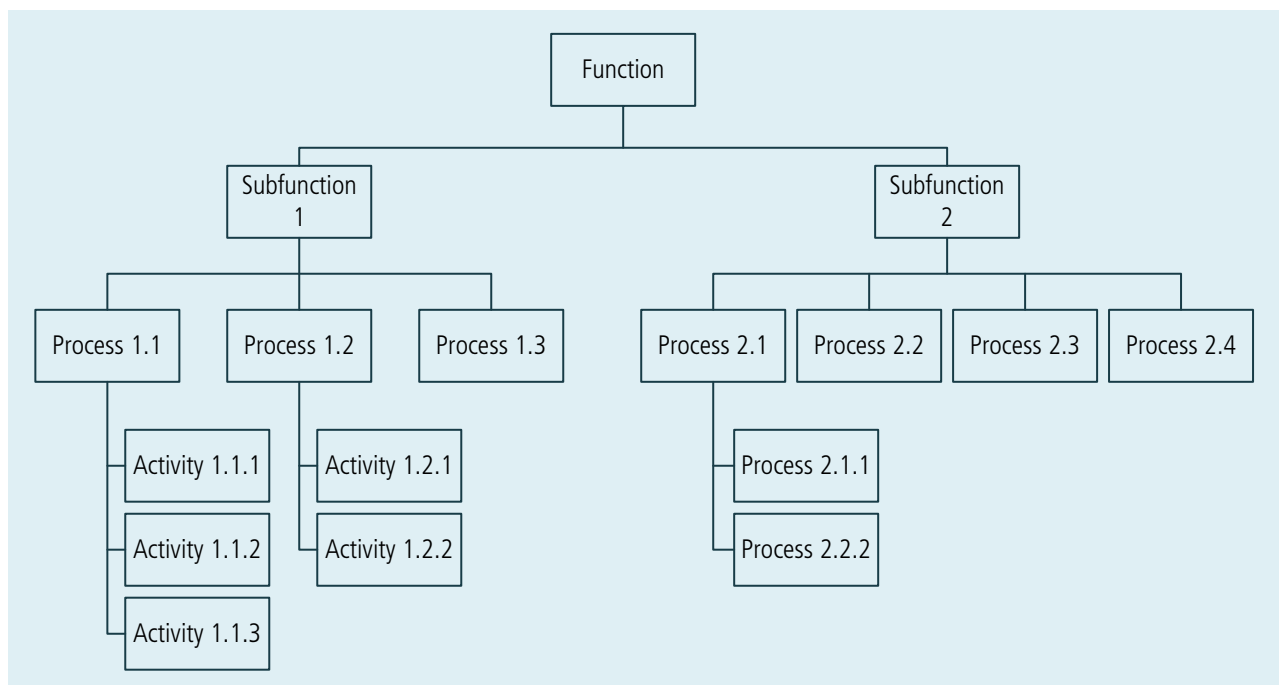
10.22.2 Description

Functional decomposition approaches the analysis of complex systems and concepts by considering them as a set of collaborating or related functions, effects, and components. This isolation helps reduce the complexity of the analysis. Breaking down larger components into sub-components allows scaling, tracking, and measuring work effort for each of them. It also facilitates evaluation of the success of each sub-component as it relates to other larger or smaller components.

The depth of decomposition may vary depending on the nature of components and objectives. Functional decomposition assumes that sub-components can and do completely describe their parent components. Any sub-component can have only one parent component when developing the functional hierarchy.

The diagram below provides an example of how a function can be broken down to manageable, measurable sub-components.

Figure 10.22.1: Functional Decomposition Diagram



10.22.3 Elements

.1 Decomposition Objectives

Objectives of functional decomposition both drive the process of decomposition and define what to decompose, how to decompose, and how deeply to decompose.

The objectives may include:

- **Measuring and Managing:** to isolate specific manageable factors that contribute to the overall result, or to identify important metrics and indicators.
- **Designing:** to simplify a design problem by reducing and isolating the object of design.
- **Analyzing:** to study the essential properties and behaviours of an artifact or phenomenon in isolation from its encompassing environment.
- **Estimating and Forecasting:** to decrease the level of uncertainty by breaking down a complex value into its constituent factors.
- **Reusing:** to create a reusable solution building block that serves a specific function for various processes.
- **Optimization:** to detect or alleviate a bottleneck, reduce function cost, or improve process quality.
- **Substitution:** to make a specific implementation of a solution component or a function easily replaceable without impacting the system as a whole.
- **Encapsulation:** combining elements to make one element.

.2 Subjects of Decomposition

Functional decomposition applies to a wide variety of versatile subjects, such as:

- **Business Outcomes:** for example, income, profit, expenses, volume of service, or volume of production.
- **Work to be Done:** this decomposition (known as a Work Breakdown Structure or WBS) breaks endeavours into phases, milestones, work activities, tasks, work items, and deliverables.
- **Business Process:** to identify its constituent parts for the purposes of measuring, managing, optimizing, or reusing the process or its components.
- **Function:** to enable its optimization or implementation.
- **Business Unit:** to enable its reverse engineering and design.
- **Solution Component:** to enable its design, implementation, or change.
- **Activity:** to enable its implementation, modification, optimization, measurement, and estimation.
- **Products and Services:** to design, implement, and improve them.

- **Decisions:** for enabling, improving, or supporting them by identifying their inputs, underlying models, dependencies, and outcomes.

.3 Level of Decomposition

The appropriate level of functional decomposition defines where, why, and when to stop decomposing the subject in order to meet the analysis objectives. The process of functional decomposition continues until the business analyst has just enough understanding and detail to proceed and can apply the results of decomposition in the execution of other tasks.

.4 Representation of Decomposition Results

Representations of functional decomposition results allow business analysts to both validate and verify the results and to use them to solve other tasks. The results can be expressed as a combination of plain textual descriptions, hierarchical lists, descriptions using special formal notations (for example, mathematical formulas, Business Process Execution Language, or programming languages), and visual diagrams. A wide variety of diagramming techniques can be used to represent functional decomposition, including:

- **Tree diagrams:** represent hierarchical partitioning of work, activities, or deliverables.
- **Nested diagrams:** illustrate hierarchical part-to-whole relationships between decomposition results.
- **Use Case diagrams:** represent decomposition of a higher-level use case.
- **Flow diagrams:** depict results of a process or function decomposition.
- **State Transition diagrams:** explain the behaviour of an object inside its composite state.
- **Cause-Effect diagrams:** elaborate on events, conditions, activities, and effects involved in producing a complex outcome or phenomenon.
- **Decision Trees:** detail the structure of a complex decision and its potential outcomes.
- **Mind Maps:** represent information in categories.
- **Component diagram:** depicts how components are wired together to form larger components and/or software systems.
- **Decision Model and Notation:** is used to analyze the business logic to ensure that it has inferential and business integrity.

10.22.4 Usage Considerations

.1 Strengths

- Makes complex endeavours possible by breaking down complex problems into feasible parts.

- Provides a structured approach to building a shared understanding of complex matters among a diverse group of stakeholders.
- Simplifies measurement and estimation of the amount of work involved in pursuing a course of action, defining scope of work, and defining process metrics and indicators.

.2 Limitations

- Missing or incorrect information at the time decomposition is performed may later cause a need to revise the results of decomposition partially or entirely.
- Many systems cannot be fully represented by simple hierarchical relationships between components because the interactions between components cause emergent characteristics and behaviours.
- Every complex subject allows multiple alternative decompositions. Exploring all alternatives can be a challenging and time-consuming task, while sticking with a single alternative may disregard important opportunities and result in a sub-optimal solution.
- Performing functional decomposition may involve deep knowledge of the subject and extensive collaboration with diverse stakeholders.

10.23

Glossary

10.23.1

Purpose

A glossary defines key terms relevant to a business domain.

10.23.2

Description

Glossaries are used to provide a common understanding of terms that are used by stakeholders. A term may have different meanings for any two people. A list of terms and established definitions provides a common language that can be used to communicate and exchange ideas. A glossary is organized and continuously accessible to all stakeholders.

10.23.3

Elements

A glossary is a list of terms in a particular domain with definitions for those terms and their common synonyms. Organizations or industries may use a term differently than how it is generally understood.

A term is included in the glossary when:

- the term is unique to a domain,
- there are multiple definitions for the term,
- the definition implied is outside of the term's common use, or
- there is a reasonable chance of misunderstanding.

The creation of a glossary should take place in the early stages of a project in order to facilitate knowledge transfer and understanding. A point of contact responsible for the maintenance and distribution of the glossary throughout the initiative is identified. Organizations that maintain glossaries often find additional uses for this information and are able to leverage the glossary for future initiatives.

Consider the following when developing a glossary:

- definitions should be clear, concise, and brief,
- acronyms should be spelled out if used in a definition,
- stakeholders should have easy and reliable access to glossaries, and
- the editing of a glossary should be limited to specific stakeholders.

10.23.4 Usage Considerations

.1 Strengths

- A glossary promotes common understanding of the business domain and better communication among all stakeholders.
- Capturing the definitions as part of an enterprise's documentation provides a single reference and encourages consistency.
- Simplifies the writing and maintenance of other business analysis information including but not limited to requirements, business rules, and change strategy.

.2 Limitations

- A glossary requires an owner to perform timely maintenance, otherwise it becomes outdated and may be ignored.
- It may be challenging for different stakeholders to agree on a single definition for a term.

10.24 Interface Analysis

10.24.1 Purpose

Interface analysis is used to identify where, what, why, when, how, and for whom information is exchanged between solution components or across solution boundaries.

10.24.2 Description

An interface is a connection between two components or solutions. Most solutions require one or more interfaces to exchange information with other solution components, organizational units, or business processes.

Interface types include:

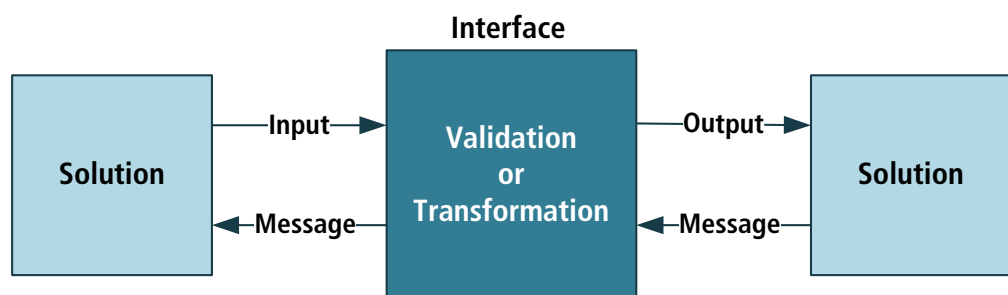
- user interfaces, including human users directly interacting with the solution within the organization,
- people external to the solution such as stakeholders or regulators,
- business processes,
- data interfaces between systems,
- application programming interfaces (APIs), and
- any hardware devices.

Interface analysis defines and clarifies the following:

- who will use the interface,
- what information is being exchanged through the interface, as well as the volume of the data,
- when information will be exchanged and how frequently,
- where the information exchange will occur,
- why the interface is needed, and
- how the interface is or should be implemented.

The early identification of interfaces allows the business analyst to provide the context for eliciting more detailed stakeholder requirements, thus determining adequate functional coverage of the solution to meet stakeholder needs. Early identification of interfaces reveals which stakeholders will benefit from or depend on the various components of the solution, which can help the business analyst determine which stakeholders should be present for other elicitation techniques.

Figure 10.24.1: Interface Analysis



10.24.3

Elements

.1 Preparing for Identification

The business analyst can leverage other techniques, such as document analysis, observation, scope modelling, and interviews, in order to understand which interfaces need to be identified. A context diagram can reveal high-level

interfaces between human actors, organizational units, business processes, or other solution components. The results of this analysis can reveal how frequently any existing interfaces are being used and any problems with them that may strengthen the case for change. The results may also help identify any key issues that need to be resolved in order for an interface solution to be created.

.2 Conduct Interface Identification

Business analysts identify what interfaces are needed in the future state for each stakeholder or system that interacts with the system. The relationship between stakeholders and interfaces can be many-to-many or, in some cases, one-to-one. Some interfaces may be less obvious or less frequent such as an interface used for regulatory functions or auditing, or for employee training. Identified interfaces can include interfaces from solutions other than the operational solution.

For each interface, business analysts:

- describe the function of the interface,
- assess the frequency of the interface usage,
- evaluate which type of interface may be appropriate, and
- elicit initial details about the interface.

.3 Define Interfaces

Requirements for an interface are primarily focused on describing the inputs to and outputs from that interface, any validation rules that govern those inputs and outputs, and events that might trigger interactions. There may be a large number of possible interaction types, each of which needs to be specified. Interactions may be triggered by the typical or alternate flow of inputs and outputs in the business solution, or by exceptional events such as failures.

Business analysts consider who will use the interface, what information is passed over the interface, and when and where the interface takes place. The interface defines user workflow between systems, user roles and privileges, and any management objectives for the interface. Interface definition is dependent upon usability guidelines, such as accessibility requirements or general workflow requirements.

In order to identify any major design issues, interfaces between solution or process components and people require detailed analysis of the interface to be conducted upfront. Interface definition includes:

- the name of the interface,
- the coverage or span of the interface,
- the exchange method between the two entities,
- the message format, and
- the exchange frequency.

10.24.4 Usage Considerations

.1 Strengths

- By engaging in interface analysis early on, increased functional coverage is provided.
- Clear specification of the interfaces provides a structured means of allocating requirements, business rules, and constraints to the solution.
- Due to its broad application, it avoids over analysis of fine detail.

.2 Limitations

- Does not provide insight into other aspects of the solution since the analysis does not assess the internal components.

10.25 Interviews

10.25.1 Purpose

An interview is a systematic approach designed to elicit business analysis information from a person or group of people by talking to the interviewee(s), asking relevant questions, and documenting the responses. The interview can also be used for establishing relationships and building trust between business analysts and stakeholders in order to increase stakeholder involvement or build support for a proposed solution.

10.25.2 Description

The interview is a common technique for eliciting requirements. It involves direct communication with individuals or groups of people who are part of an initiative.

In an interview, the interviewer directs questions to stakeholders in order to obtain information. One-on-one interviews are the most common. In a group interview (with more than one interviewee in attendance), the interviewer is careful to elicit responses from each participant.

There are two basic types of interviews used to elicit business analysis information:

- **Structured Interview:** in which the interviewer has a predefined set of questions.
- **Unstructured Interview:** in which the interviewer does not have a predetermined format or order of questions. Questions may vary based on interviewee responses and interactions.

In practice, business analysts may use a combination of the two types by adding, dropping, and varying the order of questions as needed.

Successful interviewing depends on factors such as:

- level of understanding of the domain by the interviewer,
- experience of the interviewer in conducting interviews,
- skill of the interviewer in documenting discussions,
- readiness of the interviewee to provide the relevant information and the interviewer to conduct the interview,
- degree of clarity in the interviewee's mind about the goal of the interview, and
- rapport of the interviewer with the interviewee.

10.25.3

Elements

.1 Interview Goal

When planning interviews, business analysts consider:

- the overall purpose of performing a set of interviews, based on a business need, and
- the individual goals for each interview, based on what the interviewee can provide.

The goals are to be clearly expressed and communicated to each interviewee.

.2 Potential Interviewees

Potential interviewees are identified with the help of the project manager, project sponsors, and other stakeholders, based on the goals for the interview.

.3 Interview Questions

Interview questions are designed according to the interview goals, such as:

- collecting data,
- researching the stakeholder's view of the change or proposed solution,
- developing a proposed solution, or
- building rapport with or support for the proposed solution from the interviewee.

Open-ended questions are used to elicit a dialogue or series of steps and cannot be answered in a yes or no fashion. Open-ended questions are a good tool to allow the interviewee to provide information of which the interviewer may be unaware.

Closed questions are used to elicit a single response such as yes, no, or a specific number. Closed questions can be used to clarify or confirm a previous answer.

The interview questions are often organized based on priority and significance. Examples of question order include general to specific, start to finish, and detailed to summary. Questions can also be organized based on factors such as the interviewee's level of knowledge and the subject of the interview.

Interview questions may be customized when the purpose of the interview is to gather information that is unique to the perspective of the interviewee. Standardized questions may be used when the interview results will be summarized and analyzed, such as when interview results will be tallied using a check sheet.

Interview questions can be compiled in an interview guide, which includes the interview questions, proposed timing, and follow-up questions. This will all be based on the interview type, according to the interview goals, mode of communication, and duration. The interview guide can be a document where the interviewee's responses are easily recorded. The interview guide should identify which interview questions may be omitted based upon time constraints.

.4 Interview Logistics

Ensuring a successful interview requires attention to logistics that include:

- The location for the interview. The interview is adapted to the schedule and availability of the interviewee and the mode of communication (in-person, phone, or online conferencing).
- Whether or not to record the interview, which may require the use of a scribe.
- Whether or not to send the questions to the interviewees in advance. Sending questions in advance is advisable only when the interviewee needs to collect information to prepare for the interview.
- Whether the interview results will be confidential and, if so, how the results will be summarized to avoid identifying individual interviewees.

.5 Interview Flow

Opening the interview includes:

- describing the purpose of the interview, including why the interviewees' time is needed,
- confirming the interviewees' roles and addressing any initial concerns raised by the interviewees, and
- explaining how information from the interview will be recorded and shared with the interviewees and other stakeholders during the project.

During the interview, the interviewer:

- maintains focus on the established goals and predefined questions, and adapts based upon the information provided and non-verbal communication from the interviewees,
- considers both the willingness of the interviewees to participate in the interview and to provide the required information,
- considers that several meetings might be required to conduct the entire interview,
- manages concerns raised by the interviewees by addressing them during the interview or documenting them for follow-up,

- practices active listening to confirm what the interviewer has said, and
- takes written notes or records the interview as appropriate.

Closing the interview includes:

- asking the interviewees for areas that may have been overlooked in the session,
- providing contact information for the interviewees to follow up with additional information after the meeting as needed,
- summarizing the session,
- outlining the process for how the interview results will be used, and
- thanking the interviewees for their time.

.6 Interview Follow-Up

It is important for the interviewer to organize the information and confirm results with the interviewees as soon as possible after the interview. Sharing the information that has been learned allows the interviewees to point out any missed or incorrectly recorded items.

10.25.4 Usage Considerations

.1 Strengths

- Encourages participation by and establishes rapport with stakeholders.
- Simple, direct technique that can be used in a variety of situations.
- Allows the interviewer and participant to have full discussions and explanations of the questions and answers.
- Enables observations of non-verbal behaviour.
- The interviewer can ask follow-up and probing questions to confirm their own understanding.
- Maintains focus through the use of clear objectives for the interview that are agreed upon by all participants and can be met in the time allotted.
- Allows interviewees to express opinions in private that they may be reluctant to express in public, especially when interview results are kept confidential.

.2 Limitations

- Significant time is required to plan for and conduct interviews.
- Requires considerable commitment and involvement of the participants.
- Training is required to conduct effective interviews.
- Based on the level of clarity provided during the interview, the resulting documentation may be subject to the interviewer's interpretation.
- There is a risk of unintentionally leading the interviewee.

10.26 Item Tracking

10.26.1 Purpose

Item tracking is used to capture and assign responsibility for issues and stakeholder concerns that pose an impact to the solution.

10.26.2 Description

Item tracking is an organized approach used by business analysts to address stakeholder concerns. Stakeholders may identify such item types as actions, assumptions, constraints, dependencies, defects, enhancements, and issues.

When a stakeholder concern is first raised, it is assessed to determine if it is viable. If viable, the concern is classified as a specific item type so that it can be better tracked and controlled by a process that works towards the item's closure. During its life cycle, an item is assigned to one or more stakeholders who are responsible for its resolution.

Item tracking tracks the item from the initial recording of the concern and its degree of impact to its agreed-upon closure. The item tracking record may be shared with stakeholders to ensure transparency and visibility into the status and progress of items in the record.

10.26.3 Elements

.1 Item Record

Each recorded item may contain all or any of the following attributes for item tracking. These items may be recorded using various software applications or manually catalogued for sharing between an agreed set of stakeholders.

- **Item Identifier:** a unique identifier that distinguishes one item from another.
- **Summary:** a brief description of the item.
- **Category:** a grouping of items with similar properties.
- **Type:** the kind of item raised.
- **Date Identified:** the date the item was raised as a concern.
- **Identified By:** the person who initially raised the concern.
- **Impact:** the possible consequences if the item is not resolved by the resolution due date. Impact can be assessed in relation to the initiative's time, cost, scope, or quality.
- **Priority:** the importance of this item to the impacted stakeholders.
- **Resolution Date:** the date by which the item must be resolved (or closed).
- **Owner:** the stakeholder assigned to manage the item to its closure.

- **Resolver:** the stakeholder assigned to resolve the item.
- **Agreed Strategy:** agreed-upon strategy for the item. Examples include accept, pursue, ignore, mitigate, and avoid.
- **Status:** the current status of the item within its life cycle. Examples include open, assigned, resolved, and cancelled.
- **Resolution Updates:** a running log of details about how the item's resolution is proceeding towards closure, as well as approval of its completion.
- **Escalation Matrix:** a level of escalation in case the item is not resolved by the given due date.

.2 Item Management

Each item's resolution is undertaken as prescribed by stakeholder needs and according to any organizational process standards. In some cases, one item may cause another item to be recorded and tracked. In these situations, close attention is needed so that item resolution efforts are not duplicated and are progressing in coordination. Each item must be tracked to its closure or resolution.

.3 Metrics

All stakeholders benefit from the detailed information that is maintained about any item and its progress. These items can be looked at individually for resolution or even used to define key performance indicators tailored to the item tracking process.

By reviewing this output, stakeholders can determine how well:

- items are being resolved by the proper resources,
- the initiative is progressing, and
- the item tracking process is being utilized.

10.26.4

Usage Considerations

.1 Strengths

- Ensures concerns around stakeholder requirements are captured, tracked, and resolved to the stakeholder's satisfaction.
- Allows stakeholders to rank the importance of outstanding items.

.2 Limitations

- If not careful, the copious recording of data about items may outweigh any benefits realized.
- It may use time that could be better spent on other efforts and stakeholders could become mired in details and statistics.

10.27 Lessons Learned

10.27.1 Purpose

The purpose of the lessons learned process is to compile and document successes, opportunities for improvement, failures, and recommendations for improving the performance of future projects or project phases.

10.27.2 Description

A lessons learned session (also known as a retrospective) helps identify either changes to business analysis processes and deliverables or successes that can be incorporated into future work. These techniques can also be beneficial at the close of any milestone within the effort.

Lessons learned sessions can include any format or venue that is acceptable to the key stakeholders and can be either formal facilitated meetings with set agendas and meeting roles or informal working sessions. If there are noteworthy successes, a celebration may be included in a lessons learned session.

10.27.3 Elements

Sessions can include a review of:

- business analysis activities or deliverables,
- the final solution, service, or product,
- automation or technology that was introduced or eliminated,
- impact to organizational processes,
- performance expectations and results,
- positive or negative variances,
- root causes impacting performance results, and
- recommendations for behavioural approaches.

10.27.4 Usage Considerations

.1 Strengths

- Useful in identifying opportunities or areas of improvement.
- Assists in building team morale after a difficult period.
- Reinforces positive experiences and successes.
- Reduces risks for future actions.
- Provides tangible value or metrics as a result of the effort.
- Recognizes strengths or shortcomings with the project structure, methodology, or tools that were used.

.2 Limitations

- Honest discussion may not occur if participants try to assign blame during these sessions.
- Participants may be reluctant to document and discuss problems.
- Proactive facilitation may be required to ensure that the discussions remain focused on solutions and improvement opportunities.

10.28 Metrics and Key Performance Indicators (KPIs)

10.28.1 Purpose

Metrics and key performance indicators measure the performance of solutions, solution components, and other matters of interest to stakeholders.

10.28.2 Description

A metric is a quantifiable level of an indicator that an organization uses to measure progress. An indicator identifies a specific numerical measurement that represents the degree of progress toward achieving a goal, objective, output, activity, or further input. A key performance indicator (KPI) is one that measures progress towards a strategic goal or objective. Reporting is the process of informing stakeholders of metrics or indicators in specified formats and at specified intervals.

Metrics and reporting are key components of monitoring and evaluation. Monitoring is a continuous process of data collection used to determine how well a solution has been implemented as compared to the expected results. Evaluation is the systematic and objective assessment of a solution both to determine its status and effectiveness in meeting objectives over time and to identify ways to improve the solution to better meet objectives. The top priorities of a monitoring and evaluation system are the intended goals and effects of a solution, as well as inputs, activities, and outputs.

10.28.3 Elements

.1 Indicators

An indicator displays the result of analyzing one or more specific measures for addressing a concern about a need, value, output, activity, or input in a table or graphical form. Each concern requires at least one indicator to measure it properly, but some may require several.

A good indicator has six characteristics:

- **Clear:** precise and unambiguous.
- **Relevant:** appropriate to the concern.

- **Economical:** available at reasonable cost.
- **Adequate:** provides a sufficient basis on which to assess performance.
- **Quantifiable:** can be independently validated.
- **Trustworthy and Credible:** based on evidence and research.

In addition to these characteristics, stakeholder interests are also important. Certain indicators may help stakeholders perform or improve more than others. Over time, weaknesses in some indicators can be identified and improved.

Not all factors can be measured directly. Proxies can be used when data for direct indicators are not available or when it is not feasible to collect at regular intervals. For example, in the absence of a survey of client satisfaction, an organization might use the proportion of all contracts renewed as an indicator.

When establishing an indicator, business analysts will consider its source, method of collection, collector, and the cost, frequency, and difficulty of collection. Secondary sources of data may be the most economical, but to meet the other characteristics of a good indicator, primary research such as surveys, interviews, or direct observations may be necessary. The method of data collection is the key driver of a monitoring, evaluation, and reporting system's cost.

.2 Metrics

Metrics are quantifiable levels of indicators that are measured at a specified point in time. A target metric is the objective to be reached within a specified period. In setting a metric for an indicator, it is important to have a clear understanding of the baseline starting point, resources that can be devoted to improving the factors covered by the indicator, and political concerns.

A metric can be a specific point, a threshold, or a range. A range can be useful if the indicator is new. Depending on the need, the scope of time to reach the target metric can be multi-year, annual, quarterly, or even more frequent.

.3 Structure

Establishing a monitoring and evaluation system requires a data collection procedure, a data analysis procedure, a reporting procedure, and the collection of baseline data. The data collection procedure covers units of analysis, sampling procedures, data collection instruments to use, collection frequency, and responsibility for collection. The analysis method specifies both the procedures for conducting the analysis and the data consumer, who may have strong interests in how the analysis is conducted. The reporting procedure covers the report templates, recipients, frequency, and means of communication. Baseline information is the data provided immediately before or at the beginning of a period to measure. Baseline data is used both to learn about recent performance and to measure progress from that point forward. It needs to be collected, analyzed, and reported for each indicator.

There are three key factors in assessing the quality of indicators and their metrics: reliability, validity, and timeliness. Reliability is the extent to which the data collection approach is stable and consistent across time and space. Validity is the

extent to which data clearly and directly measures the performance the organization intends to measure. Timeliness is the fit of the frequency and latency of data to the management's need.

.4 Reporting

Typically, reports compare the baseline, current metrics, and target metrics with calculations of the differences presented in both absolute and relative terms. In most situations, trends are more credible and important than absolute metrics. Visual presentations tend to be more effective than tables, particularly when using qualitative text to explain the data.

10.28.4 Usage Considerations

.1 Strengths

- Establishing a monitoring and evaluation system allows stakeholders to understand the extent to which a solution meets an objective, as well as how effective the inputs and activities of developing the solution (outputs) were.
- Indicators, metrics, and reporting also facilitate organizational alignment, linking goals to objectives, supporting solutions, underlying tasks, and resources.

.2 Limitations

- Gathering excessive amounts of data beyond what is needed will result in unnecessary expense in collecting, analyzing, and reporting. It will also distract project members from other responsibilities. On Agile projects, this will be particularly relevant.
- A bureaucratic metrics program fails from collecting too much data and not generating useful reports that will allow timely action. Those charged with collecting metric data must be given feedback to understand how their actions are affecting the quality of the project results.
- When metrics are used to assess performance, the individuals being measured are likely to act to increase their performance on those metrics, even if this causes sub-optimal performance on other activities.

10.29 Mind Mapping

10.29.1 Purpose

Mind mapping is used to articulate and capture thoughts, ideas, and information.

10.29.2 Description

Mind mapping is a form of note taking that captures thoughts, ideas, and information in a non-linear diagram. Mind maps use images, words, colour, and

connected relationships to apply structure and logic to thoughts, ideas, and information. A mind map has a central main idea supported by secondary ideas (or topics), followed by as many layers of ideas (or sub-topics) as necessary to fully capture and articulate the concept. Connections are made between ideas by branches that typically have a single keyword associated with them that explain the connection.

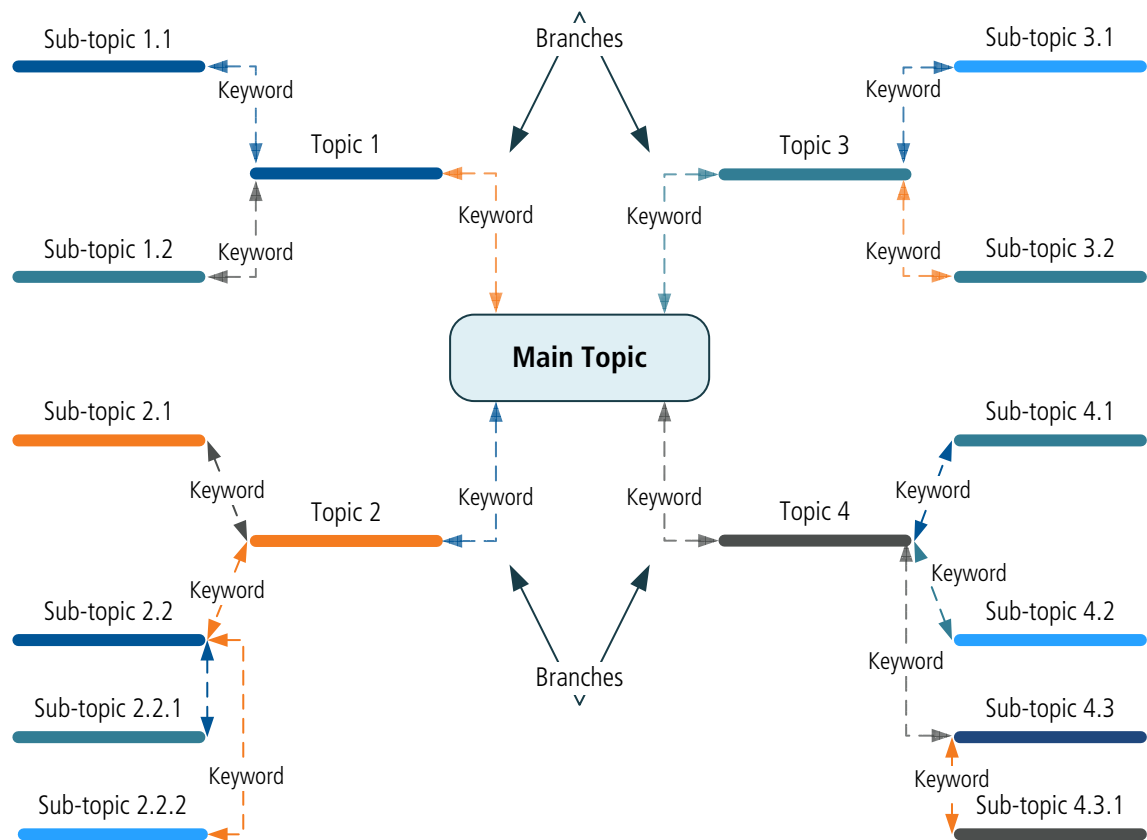
Mind maps can be developed individually or as a collaboration exercise. They can be created on paper or with the use of specialized software.

Business analysts use mind maps to:

- think through and generate ideas on complex concepts or problems,
- explore relationships between the various facets of a problem in a way that inspires creative and critical thinking, and
- present a consolidated view of complex concepts or problems.

There is no standardized format for a mind map. The intent of a mind map is to capture information in a fashion closely resembling how our minds process information. The following image is intended to illustrate the general structure and usage of mind maps.

Figure 10.29.1: The Taxonomy of a Mind Map



10.29.3**Elements****.1 Main Topic**

The main topic of a mind map is the thought or concept that is being articulated. The main topic is positioned in the centre of the images so that multiple topics and associations can branch off. Images are frequently used as the main topic because they contain a great deal of information and can be useful in stimulating associated topics.

.2 Topics

Topics are thoughts or concepts that expound upon or further articulate the main topic. Their association with the main topic is expressed through a branch (connected line) that has a keyword associated with it. There can be as many or as few topics as required to fully explore the thought or concept of the main topic.

.3 Sub-topics

Sub-topics are thoughts or concepts that expound upon or further articulate the topic and directly relate to the main topic. Their association with the topic is expressed through a branch (connected line) that has a keyword associated with it. There can be as many or as few sub-topics as required to fully explore the thought or concept of the main topic.

.4 Branches

Branches are the associations between the main topic, topics, and sub-topics. Branches include a keyword that clearly articulates the nature of the association.

.5 Keywords

Keywords are single words used to articulate the nature of the association of topics or sub-topics connected by a branch. The keywords are useful for both categorizing topics and for triggering additional associations.

.6 Colour

Colour may be used to categorize, prioritize, and analyze topics, sub-topics, and their associations. There is no defined colour coding standard for mind maps. Each mind map creator applies colour in a way that best suits their mode of thinking.

.7 Images

Images can be used in mind maps to express larger volumes of information that are unable to be expressed in short topic headings. Images are useful in stimulating creativity and innovation by generating additional thoughts, ideas, and associations.

10.29.4 Usage Considerations

.1 Strengths

- Can be used as an effective collaboration and communication tool.
- Summarizes complex thoughts, ideas, and information in a way that shows the overall structure.
- Associations and sub-topics facilitate understanding and decision making.
- Enable creative problem solving by articulating associations and generating new associations.
- Can be helpful in preparing and delivering presentations.

.2 Limitations

- Can be misused as a brainstorming tool, and the related documenting of ideas and creating associations may inhibit idea generation.
- A shared understanding of a mind map can be difficult to communicate.

10.30 Non-Functional Requirements Analysis

10.30.1 Purpose

Non-functional requirements analysis examines the requirements for a solution that define how well the functional requirements must perform. It specifies criteria that can be used to judge the operation of a system rather than specific behaviours (which are referred to as the functional requirements).

10.30.2 Description

Non-functional requirements (also known as quality attributes or quality of service requirements) are often associated with system solutions, but they also apply more broadly to both process and people aspects of solutions. They augment the functional requirements of a solution, identify constraints on those requirements, or describe quality attributes a solution must exhibit when based on those functional requirements.

Non-functional requirements are generally expressed in textual formats as declarative statements or in matrices. Declarative non-functional requirements statements will typically have a constraining factor to them. For example, errors must not exceed X per use of the process, transactions must be at least X% processed after S seconds, or the system must be available X% of the time.

10.30.3

Elements

.1 Categories of Non-Functional Requirements

Common categories of non-functional requirements include:

- **Availability:** degree to which the solution is operable and accessible when required for use, often expressed in terms of percent of time the solution is available.
- **Compatibility:** degree to which the solution operates effectively with other components in its environment, such as one process with another.
- **Functionality:** degree to which the solution functions meet user needs, including aspects of suitability, accuracy, and interoperability.
- **Maintainability:** ease with which a solution or component can be modified to correct faults, improve performance or other attributes, or adapt to a changed environment.
- **Performance Efficiency:** degree to which a solution or component performs its designated functions with minimum consumption of resources. Can be defined based on the context or period, such as high-peak, mid-peak or off-peak usage.
- **Portability:** ease with which a solution or component can be transferred from one environment to another.
- **Reliability:** ability of a solution or component to perform its required functions under stated conditions for a specified period of time, such as mean time to failure of a device.
- **Scalability:** degree with which a solution is able to grow or evolve to handle increased amounts of work.
- **Security:** aspects of a solution that protect solution content or solution components from accidental or malicious access, use, modification, destruction, or disclosure.
- **Usability:** ease with which a user can learn to use the solution.
- **Certification:** constraints on the solution that are necessary to meet certain standards or industry conventions.
- **Compliance:** regulatory, financial, or legal constraints which can vary based on the context or jurisdiction.
- **Localization:** requirements dealing with local languages, laws, currencies, cultures, spellings, and other characteristics of users, which requires attention to the context.
- **Service Level Agreements:** constraints of the organization being served by the solution that are formally agreed to by both the provider and the user of the solution.
- **Extensibility:** the ability of a solution to incorporate new functionality.

.2 Measurement of Non-Functional Requirements

Non-functional requirements often describe quality characteristics in vague terms, such as "the process must be easy to learn", or "the system must respond quickly". To be useful to developers of a solution and to be verifiable, non-functional requirements must be quantified whenever possible. Including an appropriate measure of success provides the opportunity for verification.

For example:

- "The process must be easy to learn" can be expressed as "90% of operators must be able to use the new process after no more than six hours of training", and
- "The system must respond quickly" can be expressed as "The system must provide 90% of responses in no more than two seconds".

Measurement of the other categories of non-functional requirements is guided by the source of the requirement.

For example:

- certification requirements are generally specified in measurable detail by the organization setting the standard or convention, such as ISO Certification standards,
- compliance requirements and localization requirements are set in measurable detail by their providers,
- effective service level agreements state clearly the measures of success required, and
- an organization's enterprise architecture generally defines the solution environment requirements and specifies exactly which platform or other attribute of the environment is required.

.3 Context of Non-Functional Requirements

Depending on the category of non-functional requirements, the context may have to be considered. For example, a regulatory agency may impose context-impacting compliance and security requirements, or an organization that is expanding operations abroad may have to consider localization and scalability requirements. Determining the optimal portfolio of non-functional requirements in a given organizational context is central to delivering value to stakeholders.

The assessment of a non-functional requirement, such as localization or maintainability, may impose contextual pressures on other non-functional requirements. For instance, regulations or resources in one jurisdiction may affect the maintainability of a solution in that region, and so it may justify a lower performance efficiency or reliability measure of success than in another jurisdiction.

Context is dynamic by nature and non-functional requirements may need to be adjusted or removed outright. Business analysts consider the relative stability of the context when evaluating non-functional requirements.

10.30.4**Usage Considerations****.1 Strengths**

- Clearly states the constraints that apply to a set of functional requirements.
- Provides measurable expressions of how well the functional requirements must perform, leaving it to the functional requirements to express what the solution must do or how it must behave. This will also have a strong influence on whether the solution is accepted by the users.

.2 Limitations

- The clarity and usefulness of a non-functional requirement depends on what the stakeholders know about the needs for the solution and how well they can express those needs.
- Expectations of multiple users may be quite different, and getting agreement on quality attributes may be difficult because of the users' subjective perception of quality. For example, what might be 'too fast' to one user might be 'too slow' to another.
- A set of non-functional requirements may have inherent conflicts and require negotiation. For example, some security requirements may require compromises on performance requirements.
- Overly strict requirements or constraints can add more time and cost to the solution, which may have negative impacts and weaken adoption by users.
- Many non-functional requirements are qualitative and therefore may be difficult to be measured on a scale, and may garner a degree of subjectivity by the users as to how they believe the particular requirements ultimately meet their needs.

10.31**Observation****10.31.1****Purpose**

Observation is used to elicit information by viewing and understanding activities and their context. It is used as a basis for identifying needs and opportunities, understanding a business process, setting performance standards, evaluating solution performance, or supporting training and development.

10.31.2**Description**

Observation of activities, also known as job shadowing, involves examining a work activity firsthand as it is performed. It can be conducted in either natural work environments or specially constructed laboratory conditions. The objectives of the observation dictate how it is planned for and methodically conducted.

There are two basic approaches for observation:

- **Active/Noticeable:** while observing an activity the observer asks any questions as they arise. Despite this interruption to the work flow, the observer can more quickly understand the rationale and hidden processes underlying the activity, such as decision making. A variation of this method may involve even stronger intervention into actors' activities by stimulating them to perform specific tasks. This kind of facilitated observation allows focus on the observer's objectives in order to shorten observation time or elicit specific information.
- **Passive/Unnoticeable:** during the activity the observer does not interrupt the work. Any concerns are raised once the observation is over. This allows observation of a natural flow of events without intervention by the observer, as well as measurement of the time and quality of work. A variation of this method is video recording the activity and then reviewing it with the person being observed so they may provide further clarification.

Inspection of a person's work environment helps in discovering any tools and information assets involved in performing their activities. This supports understanding of the activities, especially with the purpose of identifying needs and opportunities. This kind of observation is an important part of the technique's variation, and is known as Contextual Inquiry.

10.31.3

Elements

.1 Observation Objectives

A clear and specific objective establishes a defined purpose of the observation session.

Objectives of an observation session may include:

- understanding the activity and its elements such as tasks, tools, events, and interactions,
- identifying opportunities for improvement,
- establishing performance metrics, or
- assessing solutions and validating assumptions.

.2 Prepare for Observation

Preparing for an observation session involves planning the observation approach based on the objectives and deciding who should be viewed performing which activities at what times. While preparing for an observation session, business analysts consider the skill and experience levels of participants, the frequency of the activities being observed, and any existing documentation and analysis related to the work activity. Preparing for observation also includes creating a schedule of observations.

The plan for observation ensures that all stakeholders are aware of the purpose of the observation session, they agree on the expected outcomes, and that the session meets their expectations.

.3 Conduct the Observation Session

Before the observation session:

- explain why the observation is being conducted,
- reassure the participant that their personal performance is not being judged and that the results of this observation, among others, will be evaluated as a whole,
- inform the participant that they can stop the observation at any time, and
- recommend the sharing of any reasoning or concerns while performing the activity or soon afterwards.

During the observation session:

- attentively watch the person perform the activity and note typical and atypical tasks or steps, the manner in which any tools are used, and information content,
- record what is seen, the time taken to perform the work, its quality, any process anomalies, and the observer's own concerns or questions, and
- ask probing questions either while the work is being performed or soon after the observation session.

.4 Confirm and Present Observation Results

After the observation session, business analysts review the notes and data recorded from the observation and follow up with the participant to obtain answers to any remaining questions or to fill any gaps. Sharing these notes and data with participants may be helpful in obtaining answers to any questions or easing any concerns the participant may have.

The validated notes and data are collated with other related observations to identify similarities, differences, and trends. Findings are aggregated, summarized, and analyzed against the objectives of the session. Needs and opportunities for improvement are communicated to stakeholders.

10.31.4 Usage Considerations

.1 Strengths

- Observers can gain realistic and practical insight about the activities and their tasks within an overall process.
- Instances of informally performed tasks as well as any workarounds can be identified.
- Productivity can be viewed firsthand and realistically compared against any established performance standards or metrics.
- Recommendations for improvement are supported by objective and quantitative evidence.

.2 Limitations

- May be disruptive to the performance of the participant and the overall organization.
- Can be threatening and intrusive to the person being observed.
- While being observed, a participant may alter their work practices.
- Significant time is required to plan for and conduct observations.
- Not suitable for evaluating knowledge-based activities since these are not directly observable.

10.32 Organizational Modelling

10.32.1 Purpose

Organizational modelling is used to describe the roles, responsibilities, and reporting structures that exist within an organization and to align those structures with the organization's goals.

10.32.2 Description

An organizational model defines how an organization or organizational unit is structured. The purpose of an organizational unit is to bring together a group of people to fulfill a common purpose. The group may be organized because the people share a common set of skills and knowledge or to serve a particular market.

An organizational model is a visual representation of the organizational unit which defines:

- the boundaries of the group (who is in the group),
- the formal relationships between members (who reports to whom),
- the functional role for each person, and
- the interfaces (interaction and dependencies) between the unit and other units or stakeholders.

10.32.3 Elements

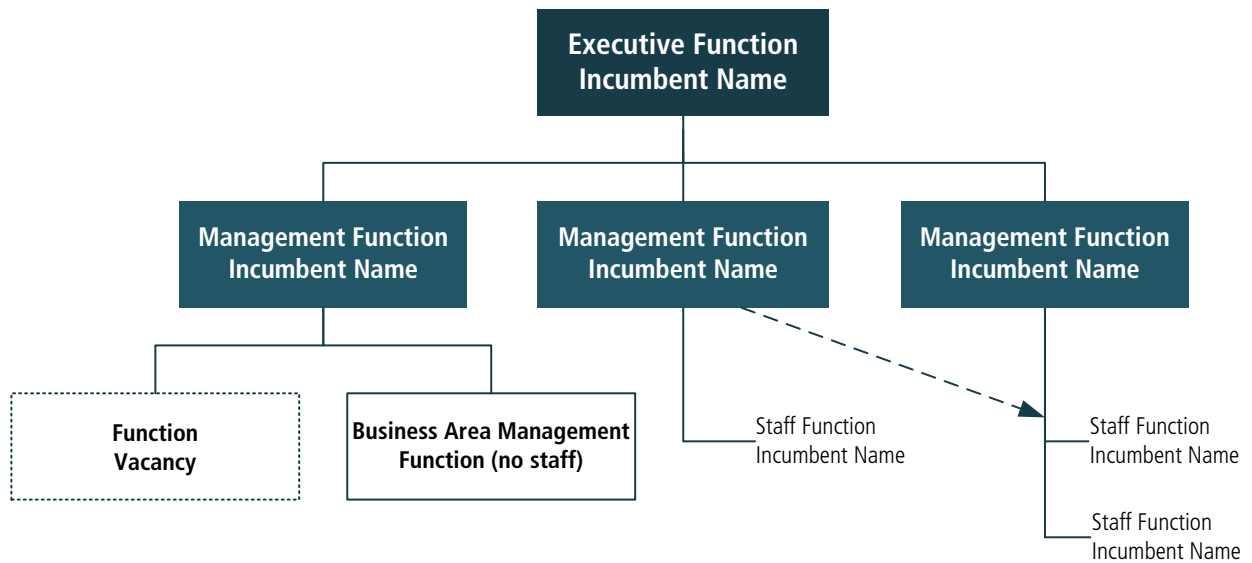
.1 Types of Organizational Models

There are three pre-eminent organizational models:

- **Functionally-oriented:** group staff together based on shared skills or areas of expertise and generally encourage a standardization of work or processes within the organization. Functional organizations are beneficial because they seem to facilitate cost management and reduce duplication of work, but are prone to develop communication and cross-functional

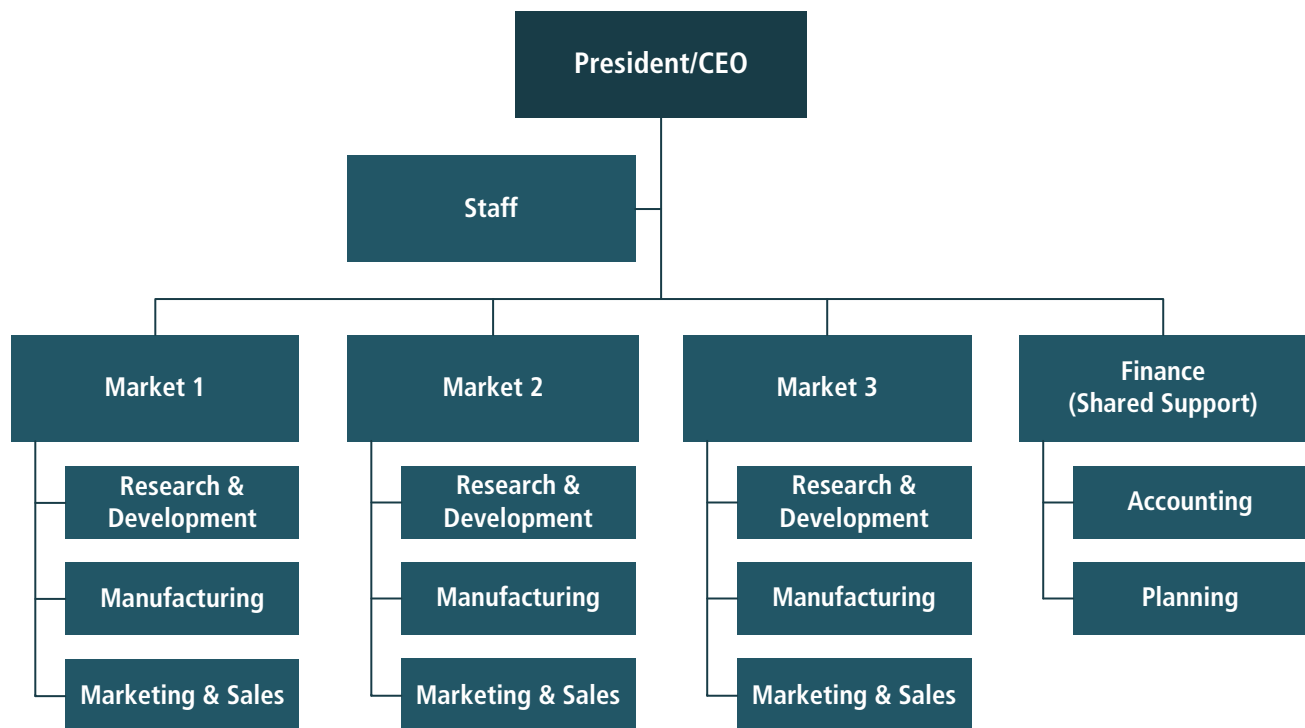
coordination problems (known informally as "silos").

Figure 10.32.1: Functionally-oriented Organizational Model



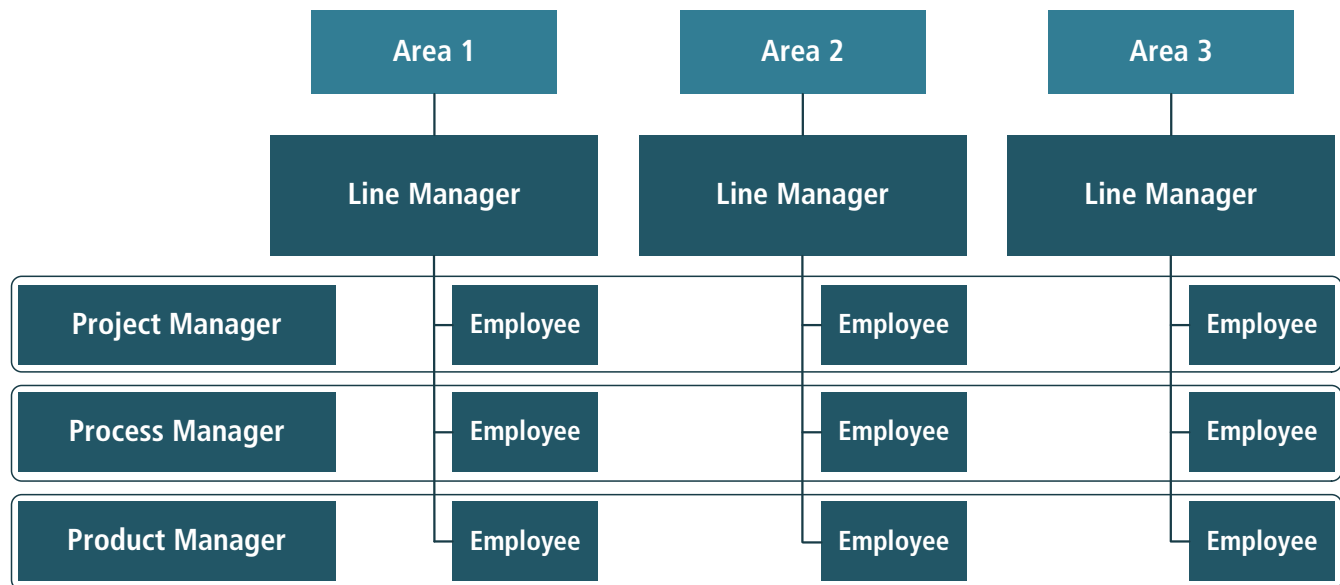
- **Market-oriented:** may be intended to serve particular customer groups, geographical areas, projects, or processes rather than grouping employees by common skills or expertise. Market-oriented structures permit the organization to meet the needs of its customers, but are prone to developing inconsistencies in how work is performed. Some may discover they are performing duplicate work in multiple areas.

Figure 10.32.2: Market-oriented Organizational Model



- **The Matrix Model:** has separate managers for each functional area and for each product, service, or customer group. Employees report to a line manager, who is responsible for the performance of a type of work and for identifying opportunities for efficiency in the work, and to a market (or product, service, or project) manager, who is responsible for managing the product or service across multiple functional areas. A challenge of the matrix model is that each employee has two managers (who are focused on different goals) and accountability is difficult to maintain.

Figure 10.32.3: Matrix Organizational Model



.2 Roles

An organizational unit includes a number of defined roles. Each role requires a certain set of skills and knowledge, has specific responsibilities, performs certain kinds of work, and has defined relationships with other roles in the organization.

.3 Interfaces

Each organizational unit has interfaces with other organizational units. Interfaces (interactions) may be in the form of communication with people in other roles and work packages that the organizational unit receives from or delivers to other units.

.4 Organizational Charts

The fundamental diagram used in organizational modelling is the organizational chart (org chart).

There is no recognized standard for org charts, although there are some conventions that most org charts follow:

- A box depicts:
 - **Organizational Unit:** people, teams, departments, or divisions. An org

chart may mix organizational units and show a mix of people, teams, and higher-level divisions.

- **Roles and People:** the roles within an organization and the people assigned to each role.
- A line depicts:
 - **Lines of Reporting:** accountability and control between units. A solid line typically denotes direct authority, while a dotted line indicates information transfer or situational authority. Lines of reporting depict the relationship between a manager and an organizational unit

.5 Influencers

Organizational charts are the primary tool for beginning organizational modelling. Organizational charts represent the formal structure of the organization. Business analysts also identify informal lines of authority, influence, and communication which may not directly align with the formal organizational chart.

Determining all of the influencers is important in planning communication and making provisions for user acceptance. One method of identifying influencers may be to ask stakeholders, “Who can I ask...” and note the answers. An influencer may be a person everyone goes to for information, direction, and advice. Another method is to note who speaks for the group in meetings.

10.32.4 Usage Considerations

.1 Strengths

- Organizational models are common in most organizations.
- Including an organizational model in business analysis information allows team members to provide support. Future projects may benefit from knowing who was involved in this project and what their role entailed.

.2 Limitations

- Organizational models are sometimes out of date.
- Informal lines of authority, influence, and communication not reflected in the org chart are more difficult to identify and may conflict with the organizational chart.

10.33 Prioritization

10.33.1 Purpose

Prioritization provides a framework for business analysts to facilitate stakeholder decisions and to understand the relative importance of business analysis information.

10.33.2

Description

Prioritization is a process used to determine the relative importance of business analysis information. The importance may be based on value, risk, difficulty of implementation, or other criteria. These priorities are used to determine which business analysis information should be targeted for further analysis, which requirements should be implemented first, or how much time or detail should be allocated to the requirements.

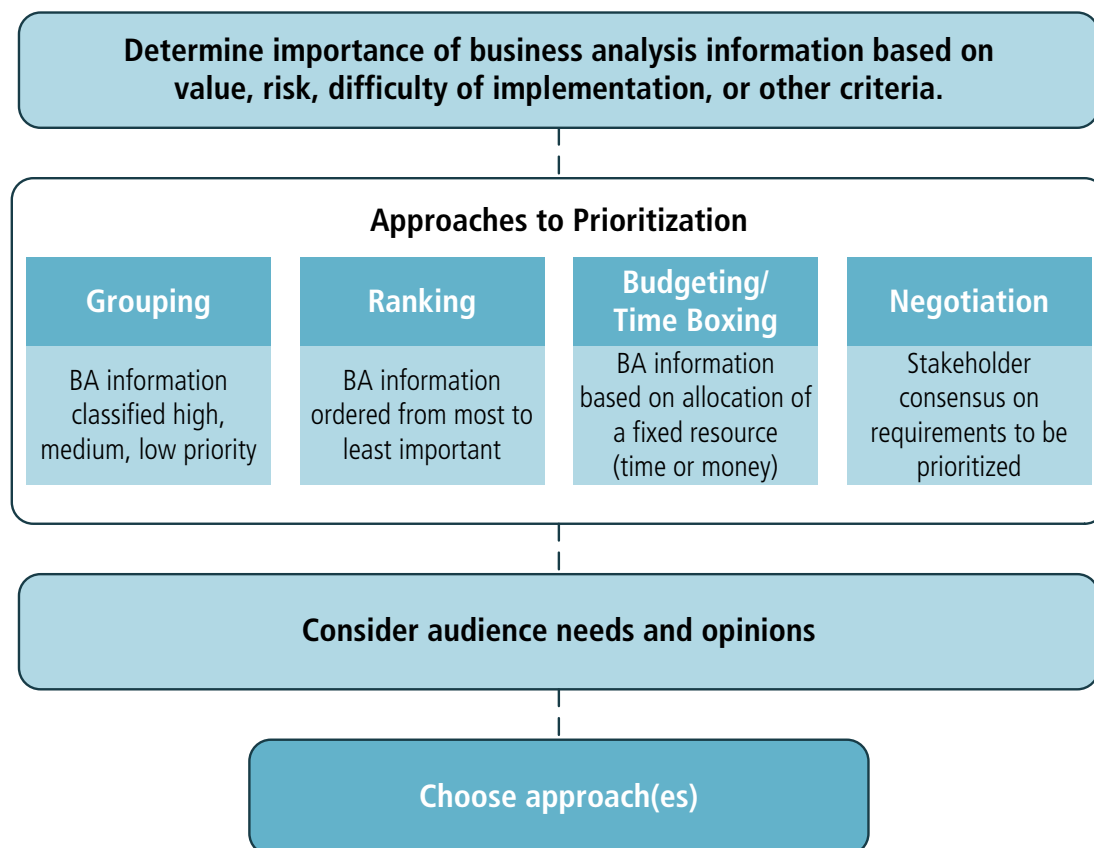
There are many approaches to prioritization. For the purpose of this technique, prioritization is classified into one of four approaches:

- Grouping,
- Ranking,
- Time boxing/Budgeting, and
- Negotiation.

When choosing a prioritization approach, business analysts consider the audience, their needs, and their opinions on the value a requirement or business analysis information brings to a stakeholder's respective area.

Business analysts revisit priorities and utilize different approaches when changes occur in the business environment, to the stakeholders, and to the business analysis information.

Figure 10.33.1: Approaches to Prioritization



10.33.3

Elements

.1 Grouping

Grouping consists of classifying business analysis information according to predefined categories such as high, medium, or low priority. Many requirements management tools support listing the priority category as an attribute of a requirement.

.2 Ranking

Ranking consists of ordering business analysis information from most to least important. Some adaptive approaches involve the explicit sequencing of requirements in an ordered list (a product backlog).

.3 Time Boxing/Budgeting

Time boxing or budgeting prioritizes business analysis information based on the allocation of a fixed resource. It is frequently used when the solution approach has been determined. Time boxing is used to prioritize requirements based on the amount of work that the project team is capable of delivering in a set period of time. Budgeting is used when the project team has been allocated a fixed amount of money. This approach is most often used when a fixed deadline must be met or for solutions that are enhanced on a regular and frequent basis.

.4 Negotiation

The negotiation approach involves establishing a consensus among stakeholders as to which requirements will be prioritized.

10.33.4

Usage Considerations

.1 Strengths

- Facilitates consensus building and trade-offs and ensures that solution value is realized and initiative timelines are met.

.2 Limitations

- Some stakeholders may attempt to avoid difficult choices and fail to recognize the necessity for making trade-offs.
- The solution team may intentionally or unintentionally try to influence the result of the prioritization process by overestimating the difficulty or complexity of implementing certain requirements.
- Metrics and key performance indicators are often not available when prioritizing business analysis information; therefore, a stakeholder's perspective of the importance may be subjective.